ISyE 7406 Project Proposal

Comparative Evaluation of Statistical Learning Methods for Autism Spectrum Disorder Risk Screening Across Age Groups

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Introduction and Motivation

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition where early identification can lead to timely clinical evaluation and intervention. I chose this topic because it is personally meaningful to me. Observing ASD-related behaviors in my son during early childhood sparked my interest in understanding how statistical learning methods can be applied to behavioral screening data. This project aims to compare statistical learning methods and identify the behavioral and demographic factors that contribute most to early ASD risk screening.

Project Description

The goal of this project is to design and compare multiple statistical learning models that classify individuals at elevated risk for Autism Spectrum Disorder based on behavioral screening responses and demographic data. It is important to note that this project does not aim to develop a diagnostic model for autism. Instead, it focuses on evaluating statistical learning models as decision-support tools for early ASD risk screening. By comparing various classification techniques across distinct age groups, my objective is to uncover which methods strike the best balance between predictive performance, sensitivity, and interpretability.

The project will include data preprocessing, exploratory data analysis, feature selection, model development, evaluation, and interpretation of the results. Particular attention will be dedicated to discerning which behavioral and demographic factors are most influential in ASD risk classification and how classifier performance varies among children, adolescents, and adults.

Dataset

This project will use publicly available Autism Spectrum Disorder (ASD) screening datasets obtained from Kaggle, originally published by Fadi Fayez Thabtah through the UCI Machine Learning Repository. The analysis will include three datasets representing children (292 observations), adolescents (104 observations), and adults (704 observations), for a total of approximately 1,100 observations. The datasets contain demographic characteristics, AQ-10 behavioral screening responses, and a binary ASD screening outcome.

Dataset:

https://www.kaggle.com/datasets/raiyangani/autistic-spectrum-disorder-screening-data

Original source:

https://archive.ics.uci.edu/datasets?skip=0&take=10&sort=desc&orderBy=NumHits&search=Autism

Research Questions & Methodology

This project aims to determine which statistical learning classifier provides the best predictive performance for Autism Spectrum Disorder (ASD) screening and whether classifier performance varies across different age groups (children, adolescents, and adults). The study will also investigate which demographic characteristics and behavioral screening questions are the strongest predictors of ASD risk, whether a reduced subset of predictors can achieve comparable performance while improving model interpretability, and which classifier provides the best balance between sensitivity and specificity in a healthcare screening setting. Additional research questions may emerge during exploratory data analysis (EDA).

The analysis will begin with data cleaning, preprocessing, and exploratory data analysis (EDA), followed by feature selection using correlation analysis, chi-square tests, and variable importance measures. Classification models will include Logistic Regression, LDA, QDA, Naive Bayes, KNN, and Random Forest. PCA will also be explored to determine whether the data can be represented using fewer principal components while retaining most of its explained variance. Model performance will be evaluated using cross-validation and multiple classification metrics, including Accuracy, Precision, Recall, Specificity, F1-score, ROC curves, and AUC.

Expected Outcomes

This project is expected to identify the statistical learning methods that provide the most reliable performance for ASD risk screening, identify the behavioral and demographic factors that contribute most to ASD risk prediction,, and provide insights into how statistical learning can support early ASD screening across different age groups.